First Attack

Computer Laboratory 2-II

Duration - 30 min

4/23/2026

- Create a folder in the **"Desktop"** using your Index number
- **e.g. EP 0000**

- Save all answers in that folder.
- Upload the resulting .zip file to the submission portal.

Question:

Create a Semester GPA Calculator that allows a student to enter their course details for a single semester. The application must calculate individual grades based on scores, determine grade points, and provide a final semester summary. You have to do,

- Design a Windows Form.
- Create input fields for 8 courses (Course Code, Credit Unit, Score).
- Implement display-only labels for "Grade" and "Points" for each row. And also apply display- only labels for “Total Credits”, “Total Points” and “GPA”.
- Add a "Semester Summary" panel showing Total Credits, Total Points, and GPA.
- Implement "Calculate GPA" and "Clear" button functionalities.

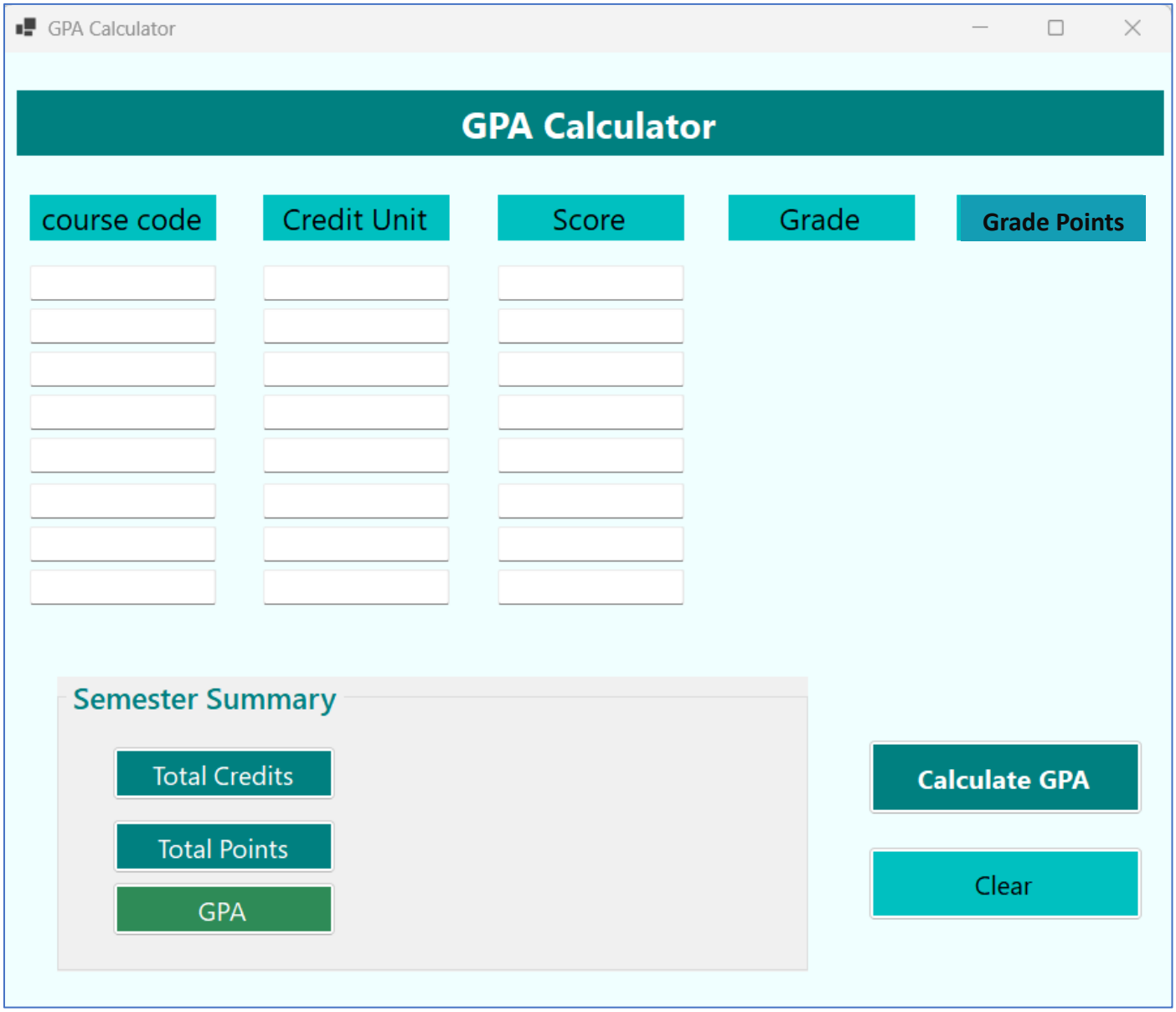
Grading Scale & Formulas

Implement the following grading system in your code.

Score Range	Grade	Grade Points
≥ 90	A+	4.0
80 – 89	A	4.0
75 – 79	A-	3.7
70 – 74	B+	3.3
65 – 69	B	3.0
60 – 64	B-	2.7
55 – 59	C+	2.3
50-54	C	2.0
45-49	C-	1.7
40-44	D+	1.3
30-39	D	1.00
≤ 29	E	0.00

Mathematical Formulas

1. Course Points = Credit Units × Grade Value
2. Semester GPA = (Total Points) / (Total Credit Units)

Functional Requirements

- Input Validation: Scores must be between 0 and 100.
- Real-time Updates: Automatically show Grades and Points when a Score is entered.
- Precision: Display GPA rounded to 2 decimal places.